Corporate Credit Early-Warning System

Zeckathon × IIT Patna TIC · **PS2: Risk & Fraud Analytics** · Team **procoder**: Mantra Khandelwal, Mohit Peshwani, Om Kumar, Omkar Mainkar

Code & data: https://github.com/mohitpeswani363/procoder-credit-ews · reproduce with `python run_v2.py`

0.73	56%	733	4/6
Hold-out average precision, 1-yr horizon (v1: 0.40)	Recall at 5% review budget (v1: 36%)	Error cost 20·FN+FP (v1: 1,069; no alerts: 1,620)	Indian defaulters ranked in the top 5% of non-defaulting controls, 4 quarters ahead

The ask and our answer

A credit desk can investigate 40 of 800 listed borrowers (5%) each quarter, and a missed default costs 20 times a false alarm. We built a reproducible early-warning system that (1) ranks borrowers with a distress model trained on the UCI Polish bankruptcy data using only ratios that can be rebuilt from Indian annual reports, (2) overlays accounting-manipulation and stress screens (Beneish M, Piotroski F, Altman Z'', interest cover, operating cash flow), (3) explains every flag with exactly-additive Shapley drivers and an investigation question, and (4) was replayed point-in-time on 11 Indian defaulters and their matched listed peers, using only accounts that were public at each cutoff.

Key findings

- **Debt-service ratios matter most.** Adding interest cover, cash-earnings-to-liabilities, short-term-liability and growth ratios lifted hold-out average precision from 0.40 to 0.73 and caught 45 instead of 29 of 81 bankruptcies inside the 5% budget, on the same frozen hold-out. The signal persists 2 and 3 years ahead (AP 0.69 / 0.66), 2.6-6x the Altman Z'' baseline.
- **Rankings transfer to India; Polish probabilities do not.** Four quarters ahead, no Indian borrower crossed the Polish-calibrated alert threshold, yet the frozen 12-ratio model placed 4 of 6 evaluable defaulters in the top 5% of listed controls four quarters before default (median percentile 98), and 3 of 4 eight quarters before. A capacity rule is a *rank* rule, so the desk should queue the top 40 by rank.
- **Explainability caught a silent failure.** Shapley drivers showed several Indian scores were driven by *missing* sales growth: in the Polish data a missing ratio signals a failing filer, in our panel it signals an extraction gap. We retrained a transfer-safe model without missing-value indicators and added a data-coverage gate instead of reporting the inflated hits.
- **Simple screens remain strong second opinions.** Altman Z'' < 1.10 flagged 3 of 6 defaulters at 4 quarters with 4 false alerts in 19 peer checks (cost 64 vs 120 for no alerts).
- **Honest limit:** only 6 of 11 defaulters have enough machine-extracted data at the 4-quarter cutoff. Every number is reproducible and every input is traceable to a PDF page; the panel is small, so Indian results are evidence, not proof.

System architecture

Stage	What happens	Code
1 Collect	196 public PDFs (179 annual reports) with original URLs and SHA-256 hashes	ews/collect.py
2 Extract	4,309 table-row candidates with page and raw line; QC rules blank implausible values (36 blanked)	ews/extract.py, panel.py
3 Point-in-time	205 firm-years for 46 issuers; public at FY-end + 6 months; originals beat restated comparatives	ews/panel.py
4 Score	25 ratios → calibrated gradient boosting (1-yr and 2-yr horizons); frozen 12-ratio ranker	ews/features.py, benchmark_v2.py
5 Screen	Altman Z'', Beneish M (8-var; proxy when COGS/SG&A absent), Piotroski F, interest cover, CFO	ews/accounting.py, replay.py
6 Explain	Permutation Shapley (exactly additive) → top-3 drivers + investigation question	ews/shapley.py
7 Queue	Rank, cap at floor(5% × N) = 40 of 800, analyst → senior officer → credit committee	ews/policy.py

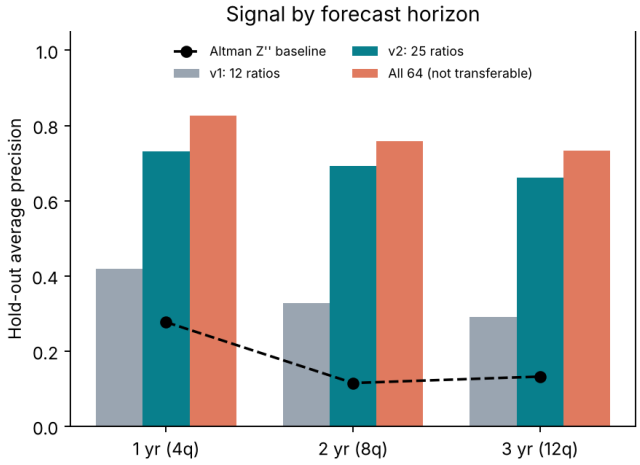
1. Distress model (UCI Polish bankruptcy benchmark)

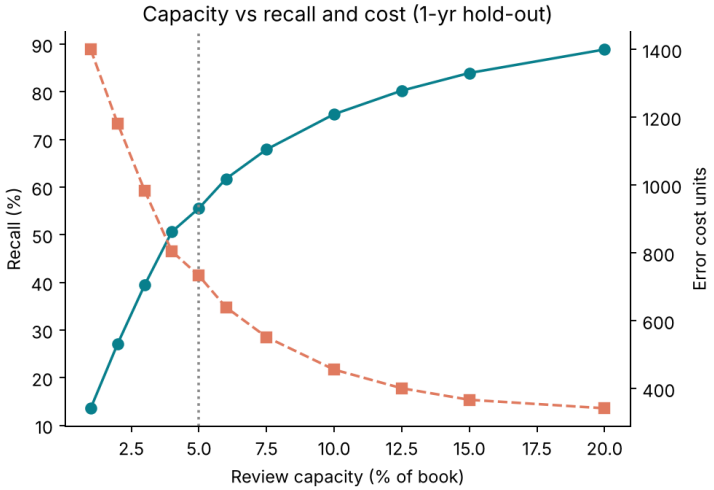
Data. UCI Polish Companies Bankruptcy (Tomczak et al.), five files of 64 ratios measured 1-5 years before the outcome. The 1-year file has 5,910 records; after removing exact duplicates and label conflicts 5,850 remain (408 bankrupt, 7.0%). **Time-aware design.** The archive has no calendar dates or company IDs, so a calendar split is impossible. We use the two time-aware tests the data allows: (i) separate models for features measured 1, 2 and 3 years before bankruptcy, and (ii) an out-of-time, out-of-domain test on Indian defaults (2013-2021) with the model trained only on Polish data. Records with identical 12-ratio vectors are grouped so no near-duplicate crosses the hold-out boundary; the 1-year hold-out is the same frozen split as v1.

Features. We kept only ratios we can rebuild from Indian Schedule III / Ind-AS statements. v1 used 12; v2 adds 13 debt-service and liquidity ratios: interest cover (X27), cash earnings / liabilities (X16, X26), pre-tax and operating profit / assets (X18, X22), sales growth (X21), cash ratio (X40), short-term liabilities / assets (X51), equity / assets and fixed assets (X10, X53), long-term liabilities / equity (X59), current assets / liabilities (X50) and cash margin (X13). **Model.** 13 prespecified configurations (logistic + histogram gradient boosting) compared by grouped 3-fold development CV; winsorising, median imputation and scaling are fitted inside folds; sigmoid calibration as a CV ensemble; the hold-out is scored once.

Hold-out (never used for choices)	Avg precision [95% CI]	ROC-AUC	Recall @5%	Precision @5%	Cost 20·FN+FP
Altman Z'' (textbook baseline)	0.28	–	21%	29%	1,321
v1 submitted model (12 ratios, frozen)	0.40	0.85	36%	50%	1,069
v2 deployed benchmark model (25 ratios)	0.73 [0.65-0.80]	0.95	56%	78%	733
All 64 ratios (upper bound, not transferable)	0.83 [0.76-0.89]	0.97	62%	86%	628
v2 transfer-safe (no missing indicators)	0.71	0.94	52%	72%	796

1,170 hold-out records, 81 bankrupt; 5% budget = 58 alerts; no-alert cost = 1,620. Hyper-parameters chosen on development CV: {'kind': 'hist_boost', 'depth': 5, 'lr': 0.03, 'iters': 400, 'leaf': 20}.





Horizon (features measured before outcome)	Hold-out n / bankrupt	v1 AP	v2 AP	v2 recall @5%	Z'' AP
1 year (≥ 4 quarters)	1,170 / 81	0.42	0.73	56%	0.28
2 years (≥ 8 quarters)	1,943 / 103	0.33	0.69	63%	0.11
3 years (≥ 12 quarters)	2,083 / 98	0.29	0.66	58%	0.13

2. Capacity, the 20:1 cost and the alert policy

For calibrated probabilities the cost-minimising rule is *alert when $p > 1/21 = 4.8\%$*. At realistic default rates that rule wants far more than 5% of the book, so **capacity, not the threshold, binds**: the development-optimal threshold (0.401) sits at the 5% cap and selects the same 58 names on the hold-out. The capacity curve shows what each extra analyst buys: going from 3% to 5% of the book adds 13 caught defaults (saving 250 cost units), and going from 5% to 10% adds another 16 at 3.2 units per extra review. Beyond about 10% each review saves less than the 20-unit value of one caught default divided by the reviews needed, which is the case for asking for a larger desk only up to roughly that point.

Step	Policy
Universe	All eligible listed borrowers at quarter-end; already-defaulted names go to workout, never silently dropped.
Data gate	Score only accounts public by the cutoff; records with fewer than 13 of 25 ratios go to manual review.
Rank	Rank by model score within the book; take floor(5% × N) = 40 of 800. Ties broken by stable ID.
Escalate	Outside the 40, any name with ≥ 2 red flags (Z'' < 1.10, interest cover < 1, M > −1.78, CFO < 0) replaces the lowest-ranked model name.

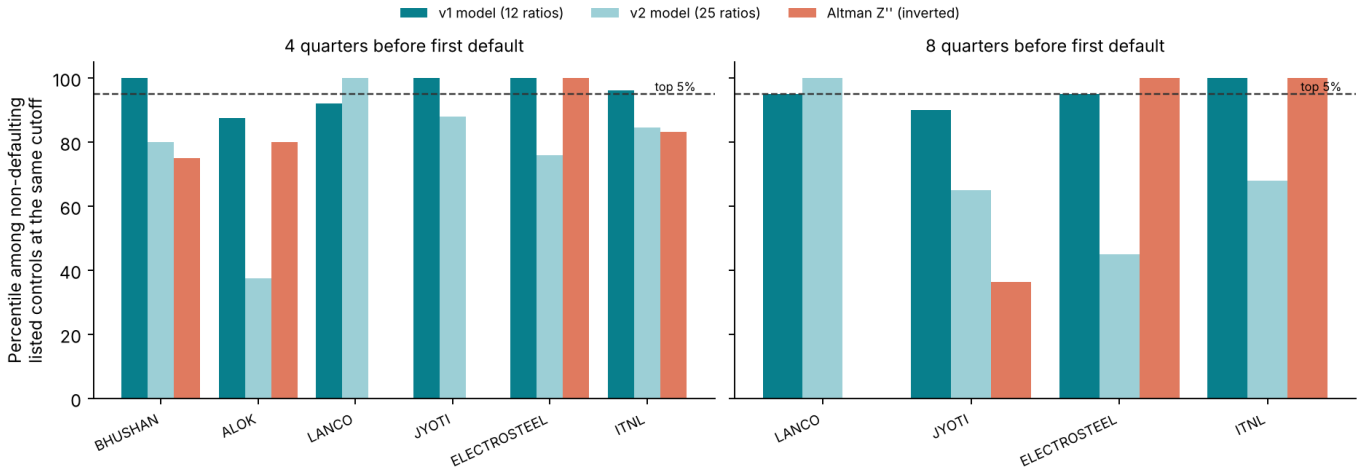
Step	Policy
Explain	Each alert gets a card: score, rank, top-3 Shapley drivers, screen values, investigation questions, source pages.
Decide	Analyst (8 × 5 names) → senior credit officer confirms/rejects → credit committee acts. Log owner, disposition, follow-up.

3. Indian case test: 11 defaulters vs matched listed peers, point-in-time

Panel. 10 companies admitted to insolvency (IBBI CIRP list, 2017-2022) plus IL&FS Transportation (ITNL) for the IL&FS group; 30 sector-matched listed controls plus 3 road-sector controls (3 earlier candidates removed for their own defaults or a late IPO). 205 firm-years from 46 issuers; 17 source-reviewed (Tier A), the rest machine-extracted with arithmetic QC (Tier B). **Event date** = first documented payment default where known (trustee notices, rating actions), otherwise CIRP admission; restructuring (CDR/SDR) is a sensitivity. **Cutoff** = last quarter-end at least 4 (or 8) quarters before the event. **No look-ahead:** a fiscal year counts only from FY-end + 6 months (Reg. 33 results in 60 days; AGM within 6 months), restated comparatives never replace originals, thresholds were fixed on Polish development data before any Indian score was seen.

Case	First default used	Cutoff 4q	Accounts used	v1 pct 4q	v2 pct 4q	Z'' 4q	v1 pct 8q
BHUSHAN	2013-12-31 (payment)	2012-12-31	2012-03-31	100%	80%	0.83	–
MONNET	2017-07-18 (CIRP)	2016-06-30	insufficient data	–	–	–	–
ALOK	2015-06-29 (payment)	2014-03-31	2012-03-31	88%	38%	2.00	–
AMTEK	2015-09-20 (payment)	2014-06-30	insufficient data	–	–	–	–
LANCO	2017-08-07 (CIRP)	2016-06-30	2015-03-31	92%	100%	–	95%
JAYPEE	2015-06-01 (payment)	2014-03-31	insufficient data	–	–	–	–
JYOTI	2017-07-04 (CIRP)	2016-06-30	2015-03-31	100%	88%	–	90%
ELECTROSTEEL	2017-07-21 (CIRP)	2016-06-30	2015-03-31	100%	76%	-1.12	95%
ERA	2015-05-01 (payment)	2014-03-31	insufficient data	–	–	–	–
FUTURE	2021-12-31 (payment)	2020-12-31	insufficient data	–	–	–	–
ITNL	2018-06-21 (payment)	2017-03-31	2016-03-31	96%	85%	0.62	100%

pct = percentile of the defaulter among all non-defaulting listed controls scorable at the same cutoff (100% = riskiest). Z'' below 1.10 is the distress zone.



4 quarters ahead (lag +6m)	Defaulters flagged	Peer false alerts	Cost 20·FN+FP	AUC vs peers
v1 model, Polish probability threshold	0 / 6	0 / 19	120 (none: 120)	0.93
v2 transfer-safe, Polish threshold	0 / 6	0 / 19	120 (none: 120)	0.82
Altman Z'' < 1.10	3 / 6	4 / 19	64 (none: 120)	0.73
Overlay: ≥ 2 red flags	2 / 6	2 / 19	82 (none: 120)	–
v2 threshold OR overlay	2 / 6	2 / 19	82 (none: 120)	–
v1 model, rank: top 5% of controls	4 / 6	by construction ≤ 5%	–	0.93

All pre-default firm-years (case years within 3 years of default = 14, peer years = 98): AUC v1 0.92, Altman Z'' 0.93 (n=62), v2 transfer-safe 0.73, interest cover 0.69. **Sensitivities:** availability lag +9 months gives 4-quarter AUC v1 0.96; counting CDR/SDR restructuring as default (Lanco Dec 2013, Monnet Dec 2015) gives 0.93.

What the Indian test says. (1) Absolute Polish probabilities are too low for Indian borrowers, so a fixed probability threshold would raise almost no alerts; ranking within the book is what transfers, and that is exactly how the 40/800 rule operates. (2) The parsimonious 12-ratio model ranks Indian defaulters better than the richer 25-ratio model, even though v2 is far stronger on the benchmark: the extra ratios are the ones our extraction most often misses (median-imputed values are not neutral for an Indian borrower). We therefore deploy v1 as the Indian

ranker, keep v2 as challenger, and name full extraction of finance costs, depreciation and prior-year sales as the change that would let v2 take over. This choice was made after seeing the panel and must be re-validated on new defaults. (3) 5 cases are not scorable 4 quarters ahead because their pre-cutoff statements were not extracted with enough fields; we report them as missing rather than guess.

4. Explanations for flagged firms (Shapley values) and investigation questions

For each defaulter and its highest-ranked control at the 4-quarter cutoff we compute permutation Shapley values of the actual calibrated model against a 32-row development background (40 antithetic permutations). Contributions sum *exactly* to the score minus the background mean (asserted in code and tests). The card turns the top drivers into the questions an analyst should ask.

Firm (role)	FY end	Score	Z''	Top-3 drivers: value (contribution)	First investigation question
ALOK (defaulter)	2012-03	0.054	2.00	Asset turnover = 0.50 (+0.023); Liabilities / assets = 0.80 (+0.013); Inventory days = 135.05 (+0.011)	Review asset utilisation and stuck/under-construction projects.
WELSPUN (control)	2013-03	0.025	–	Equity / liabilities = 0.27 (+0.010); Liabilities / assets = 0.78 (+0.009); Asset turnover = 0.86 (+0.005)	Assess the equity cushion behind creditor claims.
BHARATFORGE (control)	2013-03	0.007	4.01	Asset turnover = 0.64 (+0.007); Liabilities / assets = 0.55 (+0.001); Equity / liabilities = 0.82 (-0.000)	Review asset utilisation and stuck/under-construction projects.
BHUSHAN (defaulter)	2012-03	0.059	0.83	Quick ratio = 0.29 (+0.067); Asset turnover = 0.30 (+0.021); Equity / liabilities = 0.29 (+0.016)	Assess liquid assets against short-term obligations.
JSW (control)	2012-03	0.054	0.51	Quick ratio = 0.34 (+0.059); Asset turnover = 0.63 (+0.015); Current ratio = 0.61 (+0.006)	Assess liquid assets against short-term obligations.
ELECTROSTEEL (defaulter)	2015-03	0.109	-1.12	Quick ratio = 0.28 (+0.027); Equity / liabilities = 0.09 (+0.016); Liabilities / assets = 0.92 (+0.016)	Assess liquid assets against short-term obligations.
SSWL (control)	2015-03	0.004	–	Liabilities / assets = 0.54 (+0.001); Equity / liabilities = 0.84 (-0.000); Inventory days = 43.71 (-0.001)	Review funding dependence, creditor protection and refinancing schedule.
ITNL (defaulter)	2016-03	0.103	0.62	Quick ratio = 0.72 (+0.030); Asset turnover = 0.22 (+0.024); Retained earnings / assets = 0.06 (+0.020)	Assess liquid assets against short-term obligations.
ASHOKA (control)	2016-03	0.065	0.64	Asset turnover = 0.17 (+0.017); Liabilities / assets = 0.85 (+0.015); Equity / liabilities = 0.14 (+0.014)	Review asset utilisation and stuck/under-construction projects.
JYOTI (defaulter)	2015-03	0.079	–	Receivable days = 418.42 (+0.021); Asset turnover = 0.56 (+0.020); Liabilities / assets = 0.89 (+0.014)	Check collection delays, revenue recognition and receivable ageing.
KALPATARU (control)	2015-03	0.015	2.58	Asset turnover = 0.81 (+0.009); Liabilities / assets = 0.74 (+0.007); Equity / liabilities = 0.34 (+0.006)	Review asset utilisation and stuck/under-construction projects.
LANCO (defaulter)	2015-03	0.059	–	Quick ratio = 0.49 (+0.026); Liabilities / assets = 0.98 (+0.013); Equity / liabilities = -0.01 (+0.010)	Assess liquid assets against short-term obligations.
NTPC (control)	2014-03	0.015	3.17	Asset turnover = 0.40 (+0.008); Receivable days = 30.83 (+0.003); Liabilities / assets = 0.56 (+0.002)	Review asset utilisation and stuck/under-construction projects.
JINDALSAW (control)	2015-03	0.032	–	Asset turnover = 0.64 (+0.014); Inventory days = 112.66 (+0.011); Liabilities / assets = 0.60 (+0.003)	Review asset utilisation and stuck/under-construction projects.

Example reading. **Bhushan Steel, FY2012 accounts, cutoff Dec 2012:** the quick ratio (0.29) is the largest driver, followed by low asset turnover and thin equity relative to liabilities; Z'' was 0.83 (distress zone) and two red flags fired. The investigation question is the short-term liquidity wall, which is what failed one year later.

5. Manipulation and stress screens

Altman Z'' (6.56·WC/TA + 3.26·RE/TA + 6.72·EBIT/TA + 1.05·BVE/TL, non-manufacturer form without the +3.25 constant; distress < 1.10). **Beneish M**, 8-variable 1999 coefficients; the implementation reproduces the published Sunbeam example (−1.884). Where COGS and SG&A were not extracted we compute a proxy with GMI and SGAI held at their neutral value 1.0 and label it as such; M > −1.78 is a red flag. **Piotroski F** uses beginning-of-year assets and an explicit equity-issuance boolean; incomplete scores stay empty rather than being filled with zero. **Interest cover** < 1 and **negative operating cash flow** complete the overlay. An alert from the overlay is a reason to investigate, never proof of fraud.

Soft signals. The ledger stores dated soft evidence with its source page: related-party loans and transactions (ITNL: Rs 3,406 crore related-party loans in FY2016), subsidiary going-concern dependence and capitalised interest (Bhushan FY2012), instrument-level rating actions, and restructuring events (Lanco CDR Dec 2013, Monnet SDR Dec 2015). Panel-wide promoter-pledge (BSE shareholding patterns), auditor-resignation (LODR Reg. 30) and rating-downgrade feeds are specified in the schema but were not collected for all 44 firms in the time available, so they inform the cards and are not yet part of the measured recall.

6. Limitations

- Small Indian sample: 6 scorable defaulters at 4 quarters; confidence intervals are wide and one case changes recall by ~17 points.
- Matched controls are hand-picked large listed firms, so panel prevalence is far above a real book; we report ranks and AUC rather than portfolio precision.
- Tier-B values are machine-extracted with QC, not individually reviewed; every value links to its PDF page and raw line for audit.
- First-default dates are the earliest we could document; undiscovered earlier defaults would shorten the true lead time.
- Polish data (2000-2013, manufacturing-heavy) differs from Indian infrastructure and steel borrowers; probabilities are not Indian PDs.

7. Reproducibility and integrity

`pip install -r requirements.txt; python run_v2.py` rebuilds the benchmark (all horizons), the Indian panel, the replay, the Shapley cards, every figure and this PDF. `python run.py --test` reproduces the frozen v1 build. 76 automated tests cover formula reference values, Shapley additivity, period handling (Alok 18-month years), comparative-vs-original dating, the no-look-ahead join and fixed thresholds. Fixed seeds; frozen splits are checked on every run.

8. AI tool disclosure

Claude (Anthropic, Opus 5.5) via claude.ai: code review against the rubric; implementation of the v2 modules and tests; web look-up of three dated distress events (sources in data/curated/events.csv); drafting of this report text. All numbers are produced by the repository code; team members reviewed code, outputs and text. Full statement: docs/AI_DISCLOSURE.md.

9. Data sources and references

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[2] Zieba, M., Tomczak, S. K., Tomczak, J. M. (2016). Ensemble boosted trees with synthetic features generation in application to bankruptcy prediction. Expert Systems with Applications 58.

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[4] Insolvency and Bankruptcy Board of India: CIRP admission records (ibbi.gov.in), Agenda 7(a) 21-09-2017 and NCLT orders.

[5] Axis Trustee Services: default press release 20-Oct-2015 (axistrustee.in); ICRA and CARE rating rationales as cited in the ledgers.

[6] Business Standard (12-Dec-2013, Lanco CDR; 31-Dec-2015, Monnet SDR); BloombergQuint (Future Retail missed Rs 3,494.6 crore repayment, CARE D).

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[8] Beneish, M. D. (1999). The detection of earnings manipulation. Financial Analysts Journal 55(5); Sunbeam worked example, Kelley School.

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